Evidence 1b

- [ ] **Dual-Store Config:** Connection strings in SSM Parameter Store; credentials in Secrets Manager.
- [ ] **Centralized Logging:** App logs flowing to CloudWatch Log Group /aws/ec2/lab-rds-app.
- [ ] **CloudWatch Alarm:** Triggered on "ERROR" patterns or connection failures.
- [ ] **Incident Recovery:** Service restored by updating parameters/secrets without redeploying the instance.

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ap-northeast-1 +


```
{
    "Parameters": [
        {
            "Name": "/lab/db/endpoint1",
            "Type": "String",
            "Value": "lab-1b-host",
            "Version": 1,
            "LastModifiedDate": "2026-02-07T15:22:58.305000+00:00",
            "ARN": "arn:aws:ssm:ap-northeast-1:619076214091:parameter/lab/db/endpoint1",
            "DataType": "text"
        },
        {
            "Name": "/lab/db/name1",
            "Type": "String",
            "Value": "labdb",
            "Version": 1,
            "LastModifiedDate": "2026-02-07T14:54:55.426000+00:00",
            "ARN": "arn:aws:ssm:ap-northeast-1:619076214091:parameter/lab/db/name1",
            "DataType": "text"
        },
        {
            "Name": "/lab/db/port1",
            "Type": "String",
            "Value": "3306",
            "Version": 1,
:
```

AWS Systems Manager > Parameter Store

My parameters | Public parameters | Settings

My parameters

View details | Edit | Delete | Create parameter

Search

Name	Tier	Type	Last modified
/lab-1b/db/endpoint	Standard	String	Mon, 02 Feb 2026 19:19:40 GMT
/lab-1b/db/name	Standard	String	Mon, 02 Feb 2026 19:19:40 GMT
/lab-1b/db/port	Standard	String	Mon, 02 Feb 2026 19:19:40 GMT
/lab/db/endpoint	Standard	String	Sun, 01 Feb 2026 15:01:56 GMT
/lab/db/name	Standard	String	Sun, 01 Feb 2026 15:01:56 GMT
/lab/db/port	Standard	String	Sun, 01 Feb 2026 15:01:57 GMT

AWS Secrets Manager > Secrets > lab/rds/mysql1

Secret details

Actions

Encryption key
aws/secretsmanager

Secret name
lab/rds/mysql1

Secret ARN
arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab/rds/mysql1-JsHL6E

Secret description
Database credentials for lab RDS instance

Secret type
-

Overview | Rotation | Versions | Replication | Tags

Secret value Info

Close | Edit

Retrieve and view the secret value.

Key/value | Plaintext

Secret key	Secret value
dbname	labdb
host	lab-1b-host
password	DanBee921
port	3306
username	Dan

```
}

~ $ aws secretsmanager get-secret-value \
>   --secret-id lab/rds/mysql
{
    "ARN": "arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab/rds/mysql-dYvW2e",
    "Name": "lab/rds/mysql",
    "VersionId": "terraform-20260119210205381800000002",
    "SecretString": "{\"dbname\":\"labdb\",\"host\":\"PLACEHOLDER\",\"password\":\"StrongPassword123!\",\"port\":3306,\"username\":\"admin\"}",
    "VersionStages": [
        "AWSCURRENT"
    ],
    "CreatedDate": "2026-01-19T21:02:05.241000+00:00"
}
~ $
  $
```

```
{
    "ARN": "arn:aws:secretsmanager:ap-northeast-1:619076214091:secret:lab/rds/mysql1-JsHL6E",
    "Name": "lab/rds/mysql1",
    "VersionId": "27cf60bf-0b39-4c87-bcbe-c1732af1f1ad",
    "SecretString": "{\"dbname\":\"labdb\",\"host\":\"lab-1b-host\",\"password\":\"DanBee921\",\"port\":\"3306\",\"username\":\"Dan\"}",
    "VersionStages": [
        "AWSCURRENT"
    ],
    "CreatedDate": "2026-02-02T19:33:06.919000+00:00"
}
```

```
~ $ aws logs describe-log-groups \
>   --log-group-name-prefix /aws/ec2/lab-rds-app
{
    "logGroups": [
        {
            "logGroupName": "/aws/ec2/lab-rds-app",
            "creationTime": 1769958116703,
            "retentionInDays": 7,
            "metricFilterCount": 1,
            "arn": "arn:aws:logs:ap-northeast-1:619076214091:log-group:/aws/ec2/lab-rds-app:*",
            "storedBytes": 0,
            "logGroupClass": "STANDARD",
            "logGroupArn": "arn:aws:logs:ap-northeast-1:619076214091:log-group:/aws/ec2/lab-rds-app",
            "deletionProtectionEnabled": false
```

```
~ $
~ $ aws logs filter-log-events \
>   --log-group-name /aws/ec2/lab-rds-app \
>   --filter-pattern "ERROR"
{
    "events": [],
    "searchedLogStreams": []
```

```
{
    "MetricAlarms": [
        {
            "AlarmName": "lab-db-connection-failure",
            "AlarmArn": "arn:aws:cloudwatch:ap-northeast-1:619076214091:alarm:lab-db-connection-failure",
            "AlarmConfigurationUpdatedTimestamp": "2026-02-01T15:01:59.766000+00:00",
            "ActionsEnabled": true,
            "OKActions": [
                "arn:aws:sns:ap-northeast-1:619076214091:lab-db-incidents"
            ],
            "AlarmActions": [
                "arn:aws:sns:ap-northeast-1:619076214091:lab-db-incidents"
            ],
            "InsufficientDataActions": [],
            "StateValue": "INSUFFICIENT_DATA",
            "StateReason": "Unchecked: Initial alarm creation",
            "StateUpdatedTimestamp": "2026-02-01T15:01:59.766000+00:00",
            "MetricName": "DBConnectionErrors",
            "Namespace": "Lab/RDSApp",
            "Statistic": "Sum",
            "Dimensions": [
                {
                    "Name": "LogGroupName",
                    "Value": "/aws/ec2/lab-rds-app"
                }
            ],
            "Period": 300,
            "EvaluationPeriods": 1,
            "Threshold": 3.0,
            "ComparisonOperator": "GreaterThanOrEqualToThreshold",
            "TreatMissingData": "missing",
            "StateTransitionedTimestamp": "2026-02-01T15:01:59.766000+00:00"
        }
    ],
```

Reflection Questions (Lab 1B)

- **A)** Why might Parameter Store still exist alongside Secrets Manager in a production environment? To manage cost and so that we don't need to hardcode password sensitive information which is bad practice. Secrets manager is best for sensitive data
- **B)** What is the first thing that usually breaks during a secret rotation? The database connection

- **C)** Why should alarms be based on symptoms (e.g., 5xx errors) instead of causes (e.g., high CPU)? It allows you to observe when there are issues earlier in the journey of the application
- **D)** How does this dual-store pattern reduce Mean Time to Recovery (MTTR)? Less time was spend destroying resources and creating a new password. A
- **E)** What would you automate next? The introduction of a further alarm

Evidence 1b-incident response

A. Configuration Proof (CLI Output)

- [ ] **Parameter Store Audit:** Output of aws ssm get-parameters showing the endpoint and port.
- [ ] **Secrets Manager Audit:** Output of aws secretsmanager get-secret-value proving the app can retrieve credentials.
- [ ] **Log Group Verification:** Output of aws logs describe-log-groups showing the application log group exists.

B. Incident Response Evidence

- [ ] **Simulated Failure:** Documentation of how you triggered the error (e.g., stopping RDS, changing the secret without updating the DB, or blocking Security Groups).
- [ ] **Alarm State:** CLI output proving the CloudWatch Alarm transitioned to the ALARM state.
- [ ] **Log Analysis:** Evidence from aws logs filter-log-events --filter-pattern "ERROR" showing the explicit DB connection failure messages.
- [ ] **Recovery Confirmation:** A curl test to the EC2 application showing it resumed operation after you fixed the configuration **without redeploying the EC2 instance**.